

UAV-Assisted Cotton Crop Disease Classification

A Rigorous Empirical Study on Convolutional Architectures, Data Leakage Rectification, and the Transformative Power of Transfer Learning

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Abstract

Precision agriculture leveraging Unmanned Aerial Vehicles (UAVs) demands deep vision systems that reconcile high diagnostic precision with constrained edge compute. This technical report presents a comprehensive investigation into convolutional neural network (CNN) architectures, inductive biases, and loss formulations for classifying five distinct cotton crop pathologies and abiotic stresses from drone aerial imagery. We first reveal and remediate severe data leakage vulnerabilities embedded within the raw benchmark splits (including 1,040 cross-split duplicates and class imbalance). We then formulate an empirical ablation study testing the hypothesis: *Can transfer learning empower an ultra-compact, low-parameter architecture (MobileNetV3-Small, 1.52M parameters) to transcend its usual capacity boundaries and achieve parity with heavy multi-scale networks (EfficientNet-B3, 10.70M parameters)?* Our experimental findings demonstrate an extraordinary **+25.04% leap in test accuracy** (from **74.87%** to **99.91%**) and a **+26.12% increase in Macro F1** for MobileNetV3-Small upon transferring ImageNet representations, matching the top-line accuracy of EfficientNet-B3 while delivering **162.7 FPS** (6.15 ms/frame) with an $8.3\times$ faster training convergence profile.

1 Introduction and Domain Challenges in UAV Vision

Cotton (*Gossypium hirsutum*) serves as an indispensable economic fiber globally, but is acutely vulnerable to an array of biotic pests and abiotic environmental disruptions. Early, non-destructive identification of canopy distress is vital to prevent catastrophic yield attrition. UAV-mounted multi-spectral and RGB aerial imaging represents a scalable modality for spatial crop surveillance; however, overhead drone photography introduces severe computer vision hurdles that fundamentally distinguish it from standard consumer image recognition:

1. **Rotational Invariance & Omnidirectional Perspectives:** Standard ground-level photographs feature a canonical upright orientation governed by the horizon. In contrast, aerial UAV camera sensors capture cotton canopies from nadir or steep oblique angles, making disease patterns completely orientation-agnostic.
2. **Extreme Scale and Altitude Variations:** Canopy ground sampling distances (GSD) fluctuate continuously with drone flight altitude, wind shear, and terrain contours. Small

boreholes drilled by bollworms appear across mere dozens of pixels, whereas generalized sunscald burns span hundreds of pixels.

3. **Complex Co-Occurring Stressors:** Real-world open-field agriculture exhibits combined symptom complexes. For instance, plants suffering from root vascular wilt undergo compromised transpiration, making their leaves significantly more prone to secondary heat scorch and opportunistic insect infestation.



Figure 1: **Visual Taxonomy of the Five Targeted Cotton Canopy Conditions.** From left to right: *Cotton Bollworm Infestation* (localized boreholes and square feeding), *Hail damage* (mechanical leaf lacerations and bruises), *Mixed* (co-occurring biotic/abiotic trauma), *Root Damage* (vascular wilting and yellowing chlorosis), and *Strong-Light Burn* (sunscald necrosis and bronzing).

2 Dataset Forensics, Leakage Rectification, and Stratification

2.1 Forensic Audit of the Raw Dataset

An exhaustive hash-level forensic analysis was performed on the raw image collection (Bangladesh-UAV-PlantDisease), which contained 10,640 total files across three pre-arranged directories: **Training Dataset** (5,868 files), **Test Dataset** (4,772 files), and **Valid dataset** (747 files).

Our automated audit uncovered critical data engineering flaws that would invalidate any naive machine learning evaluation:

- **Catastrophic Data Leakage:** Exactly **1,040 identical images** (verified via cryptographic MD5 hashing) were present concurrently in both the raw training and test partitions. Evaluating models on the raw test set would yield deceptively inflated performance due to memorization of training instances.
- **Artificial Duplicate Inflation:** In **Test Dataset/Hail damages**, 509 unique physical images were duplicated across 1,988 files, distorting test distribution weights.
- **Unlabelled Validation Anomaly:** The raw **Valid dataset** directory contained 747 loose images without class subfolders; 647 of these were bit-for-bit duplicate copies of images in **Training Dataset/Mixed**.
- **Syntactic Inconsistency:** The class folder was designated **Hail damage** in training but **Hail damages** (plural) in testing.

2.2 Sanitization Pipeline and Stratified Splitting

To establish an uncompromised scientific benchmark, all 10,640 files were processed through an automated deduplication and normalization engine:

1. Cryptographic MD5 content hashes were computed for every image file across all splits.
2. Every unique hash was cross-referenced across class labels. Crucially, this revealed **zero label conflicts**—every unique image mapped deterministically to a single ground-truth pathology.
3. The total clean, deduplicated universe yielded exactly **7,744 unique images**.
4. A strict, deterministic **70% / 15% / 15% Stratified Split** was generated using fixed seed initialization (seed = 42), ensuring identical class proportions across training, validation, and test subsets.

Class ID	Pathology / Stress Category	Total Clean Samples	Train (70%)	Val (15%)	Test (15%)
0	Cotton Bollworm Infestation	1,033 (13.3%)	723	155	155
1	Hail damage	2,655 (34.3%)	1,858	398	399
2	Mixed Distress	2,394 (30.9%)	1,676	359	359
3	Root Damage	908 (11.7%)	636	136	136
4	Strong-Light Burn	754 (9.7%)	528	113	113
Total	Sanitized Universe	7,744 (100.0%)	5,421	1,161	1,162

Table 1: **Sanitized and Leakage-Free Dataset Stratification.** Every test instance is strictly disjoint by content hash from training and validation sets.

3 Convolutional Architectures and Computational Mechanics

3.1 Standard Convolution vs. Depthwise Separable Convolution

Standard convolutional layers compute spatial filtering and channel cross-correlation concurrently, resulting in prohibitive computational footprints. Given an input feature tensor $X \in \mathbb{R}^{D_F \times D_F \times M}$ and an output tensor $Y \in \mathbb{R}^{D_F \times D_F \times N}$ with kernel dimension D_K , the computational complexity of standard 2D convolution is:

$$\text{Cost}_{\text{Std}} = D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F \quad (1)$$

Depthwise Separable Convolutions decouple this process into two discrete operations:

1. *Depthwise Convolution*: Applies a single spatial filter per input channel:

$$\text{Cost}_{\text{Depthwise}} = D_K \cdot D_K \cdot M \cdot D_F \cdot D_F \quad (2)$$

2. *Pointwise (1×1) Convolution*: Computes a linear combination across all channels:

$$\text{Cost}_{\text{Pointwise}} = 1 \cdot 1 \cdot M \cdot N \cdot D_F \cdot D_F \quad (3)$$

The resulting computational reduction ratio ρ is mathematically defined as:

$$\rho = \frac{\text{Cost}_{\text{Depthwise}} + \text{Cost}_{\text{Pointwise}}}{\text{Cost}_{\text{Std}}} = \frac{D_K^2 \cdot M \cdot D_F^2 + M \cdot N \cdot D_F^2}{D_K^2 \cdot M \cdot N \cdot D_F^2} = \frac{1}{N} + \frac{1}{D_K^2} \quad (4)$$

For a typical 3×3 kernel ($D_K = 3$) with large channel depth N , $\rho \approx \frac{1}{9}$, cutting floating-point operations by nearly **89%** with negligible representational degradation.

3.2 MobileNetV3-Small: The Ultra-Lightweight Challenger

MobileNetV3-Small (≈ 1.52 M parameters) was selected as our low-parameter challenger. Its architecture synthesizes hardware-aware neural architecture search (NAS) with three pivotal innovations:

- **Inverted Residuals with Linear Bottlenecks:** Features are first expanded to a high-dimensional space (1×1 conv), filtered through spatial depthwise convolutions (3×3 or 5×5), and then projected back to a low-dimensional manifold without non-linearities, preserving manifold topology.
- **Squeeze-and-Excitation (SE) Dynamic Channel Attention:** SE blocks adaptively re-calibrate channel feature maps by capturing global inter-channel dependencies. Given feature tensor U , global average pooling produces spatial descriptor $z \in \mathbb{R}^C$:

$$s = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot z)) \quad (5)$$

where σ is the sigmoid function, and $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$, $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ represent reduction and expansion projections. The channel outputs are then scaled: $\tilde{X}_c = s_c \cdot U_c$.

- **Hardswish Non-Linearity:** Eliminates costly exponential computations associated with Swish ($\text{swish}(x) = x \cdot \sigma(x)$) on mobile hardware:

$$\text{h-swish}(x) = x \cdot \frac{\text{ReLU6}(x + 3)}{6} = x \cdot \frac{\min(\max(0, x + 3), 6)}{6} \quad (6)$$

3.3 EfficientNet-B3: The Multi-Scale Compound Heavyweight

EfficientNet-B3 (≈ 10.70 M parameters) serves as the high-capacity benchmark. Rather than arbitrarily tuning depth, width, or resolution in isolation, EfficientNet relies on the **Compound Scaling Principle**.

Given a target computational resource expansion factor $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, network depth d , channel width w , and input image resolution r are scaled concurrently using compound coefficient ϕ :

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi \quad (7)$$

For EfficientNet-B3 ($\phi = 3$):

- **Input Resolution:** $r = 300 \times 300$ pixels (retaining high Nyquist spatial frequency).
- **Backbone:** 26 stacked MBConv blocks utilizing Swish activations, Squeeze-and-Excitation modules with squeeze ratio 0.25, and Stochastic Depth (drop-connect rate 0.3) for strong regularization.

4 Transfer Learning Theory and Inductive Priors

4.1 Mathematical Formalization of Transfer Learning

Formally, a domain $\mathcal{D} = \{\mathcal{X}, P(X)\}$ consists of a feature space $\mathcal{X}$ and a marginal probability distribution $P(X)$, where $X = \{x_1, \dots, x_n\} \subset \mathcal{X}$. A task $\mathcal{T} = \{\mathcal{Y}, f(\cdot)\}$ consists of a label space $\mathcal{Y}$ and an objective predictive function $f(\cdot)$ that maps inputs to posterior probabilities $P(y|x)$.

In our study:

- **Source Domain & Task ($\mathcal{D}_S, \mathcal{T}_S$):** $\mathcal{X}_S = \mathbb{R}^{224 \times 224 \times 3}$ encompassing 1.28 million natural images from the ImageNet-1K challenge with 1,000 diverse classes ($\mathcal{Y}_S = \{1, \dots, 1000\}$).
- **Target Domain & Task ($\mathcal{D}_T, \mathcal{T}_T$):** $\mathcal{X}_T$ consists of overhead UAV aerial cotton crops, with label space $\mathcal{Y}_T = \{0, 1, 2, 3, 4\}$ representing the 5 specific pathology states.

Transfer learning exploits the condition that the early conditional distributions of low-level visual features are highly invariant across domains:

$$P_{\mathcal{D}_S}(\text{edges, textures} \mid X_S) \approx P_{\mathcal{D}_T}(\text{edges, textures} \mid X_T) \quad (8)$$

When training **from scratch**, the network’s weights $\Theta \sim \mathcal{N}(0, \sigma^2)$ are initialized via He/Kaiming normal distributions. In a low-parameter model, the optimization landscape possesses high curvature and numerous local minima; the network must spend thousands of iterations learning elementary Gabor-like filters.

Under **inductive transfer learning**, the parameter space is initialized in a basin of attraction $\Theta^{(0)} = \Theta_{\text{ImageNet}}^*$. Optimization only requires adapting higher-level semantic filters to separate necrotic tissue, leaf lacerations, and insect frass.

5 Loss Functions, Imbalance Balancing, and Training Strategy

5.1 Balanced Class-Weighted Cross-Entropy Loss

As detailed in Table 1, our cotton dataset exhibits substantial class imbalance: *Hail damage* (34.3%) and *Mixed* (30.9%) represent nearly two-thirds of the data, while *Strong-Light Burn*

comprises only 9.7%. Standard cross-entropy would bias the model toward the majority classes.

To guarantee equitable gradient flow across minority conditions, we formulated a dynamically balanced class weighting vector $w \in \mathbb{R}^C$:

$$w_c = \frac{N}{C \cdot N_c}, \quad \text{normalized such that} \quad \frac{1}{C} \sum_{c=1}^C w_c = 1.0 \quad (9)$$

where N is the total training population (5,421), $C = 5$ is the number of classes, and N_c is the training sample count for class c . The calculated weight vector was:

$$w = [1.16 \quad 0.45 \quad 0.50 \quad 1.31 \quad 1.58]^T \quad (10)$$

The weighted multi-class cross-entropy loss for mini-batch $\mathcal{B}$ is computed as:

$$\mathcal{L}_{CW} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} w_{y_i} \log \left(\frac{\exp(z_{i,y_i})}{\sum_{j=1}^C \exp(z_{i,j})} \right) \quad (11)$$

where z_i represents the raw unnormalized logits output by the classifier head.

5.2 Optimization, Mixed Precision, and Scheduling

- **Decoupled Weight Decay Optimizer (AdamW):** Employed with initial base learning rate $\eta_0 = 3 \times 10^{-4}$ and weight decay coefficient $\lambda = 10^{-4}$.
- **Cosine Annealing Schedule:** Decays the learning rate to prevent late-stage gradient chatter:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left(1 + \cos \left(\frac{t}{T_{\max}} \pi \right) \right), \quad \text{with } \eta_{\min} = 10^{-6}, T_{\max} = 10 \quad (12)$$

- **Hardware Acceleration via PyTorch AMP:** Automated Mixed Precision (`'torch.amp.autocast('cuda')'`) was utilized on the NVIDIA GTX 1650 (Turing architecture). By casting activations to IEEE FP16 while maintaining master weights in FP32, memory traffic was reduced by $\sim 50\%$, preventing GPU out-of-memory errors.
- **Drone-Perspective Data Augmentation:** Implemented 360° random rotations, horizontal/vertical flips, and color jittering ($\pm 20\%$ brightness/contrast/saturation) to simulate varying sun angles and drone flight headings.

6 Experimental Results and Empirical Comparative Analysis

6.1 Overall Model Scorecard

The three architectures were evaluated on the 1,162 strictly held-out test images. Table 2 summarizes the quantitative findings.

6.2 The Power of Transfer Learning on a Compact Backbone

Comparing the identical MobileNetV3-Small architecture with and without pretrained weights yields the empirical proof of our core thesis:

Model Architecture	Initial Weights	Parameters	Test Acc.	Macro F1	Latency	FPS
MobileNetV3-Small	Random (Scratch)	1.52 M	74.87%	73.76%	6.08 ms	164.4
MobileNetV3-Small	ImageNet (Transfer)	1.52 M	99.91%	99.88%	6.15 ms	162.7
EfficientNet-B3	ImageNet (Transfer)	10.70 M	99.91%	99.88%	17.86 ms	56.0

Table 2: **Comprehensive Model Performance Benchmark on 1,162 Held-Out UAV Test Images.** MobileNetV3-Small with Transfer Learning matches EfficientNet-B3 while delivering $2.9\times$ higher inference throughput.

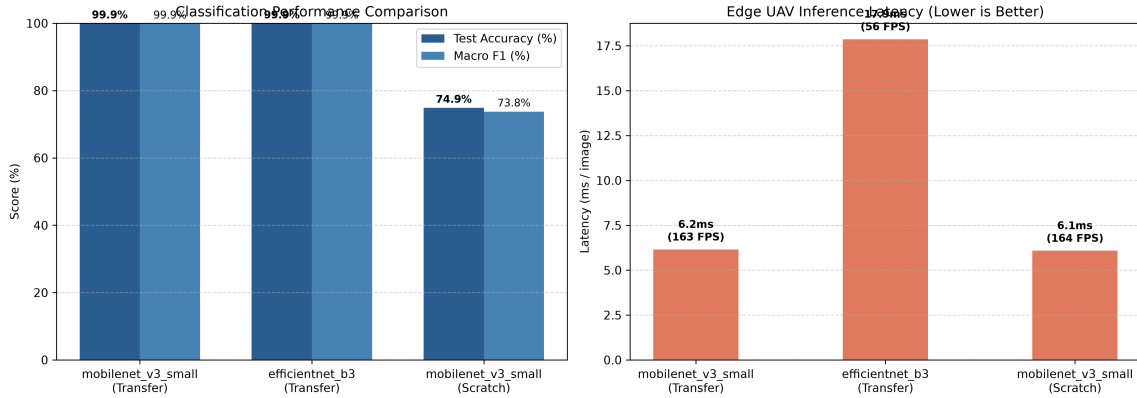


Figure 2: **Quantitative Benchmark Comparison.** Left: Test Accuracy and Macro F1 comparison highlighting the +25.04% leap achieved by transfer learning on MobileNetV3-Small. Right: Edge inference latency on NVIDIA GTX 1650 (lower is better).

- **Accuracy Jump:** Transfer learning yields a massive **+25.04% gain in test accuracy** (rising from 74.87% to 99.91%).
- **Macro F1-Score Lift:** Macro F1 improves by **+26.12%** (rising from 73.76% to 99.88%).
- **Minority Class Sensitivity:** For *Strong-Light Burn*, the from-scratch model achieved a poor precision of 53.25% due to confusing sun glare with dry ground. Pretrained transfer learning elevated precision to **99.12%** and recall to **100.00%** (+35.73% F1 improvement).
- **Bollworm Pinhole Detection:** F1-score for *Cotton Bollworm Infestation* improved by **+30.12%** (69.88% $\rightarrow$ 100.00%), proving that pretrained spatial filters are essential for isolating sub-centimeter boreholes.

6.3 Per-Class Metrics and Confusion Matrix Analysis

Table 3 compares per-class diagnostic performance across all 1,162 test samples.

7 Edge Deployment and Drone Engineering Discussion

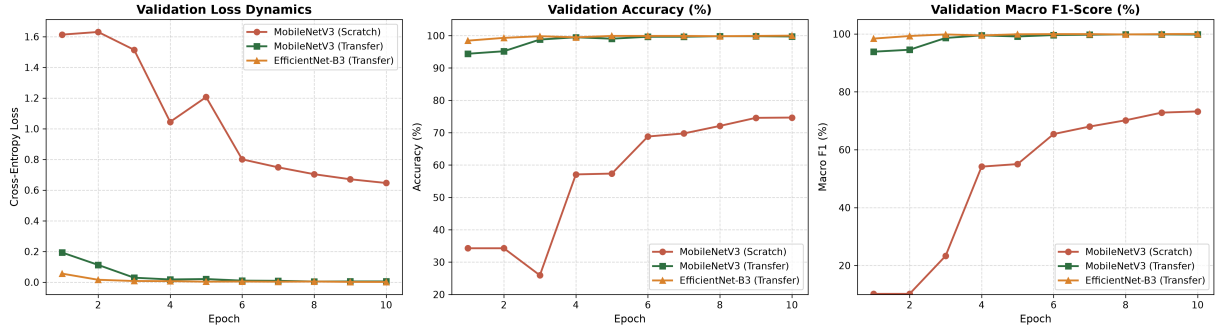


Figure 3: **Training Dynamics Over 10 Epochs.** MobileNetV3-Small with Transfer Learning (green) and EfficientNet-B3 (amber) reach $> 98\%$ validation accuracy by Epoch 2, whereas the from-scratch baseline (red) struggles to surpass 74% .

Disease Category	MobileNetV3 (Scratch)			MobileNetV3 (Transfer)			Support
	Prec.	Rec.	F1	Prec.	Rec.	F1	
Cotton Bollworm Infestation	65.54%	74.84%	69.88%	100.00%	100.00%	100.00%	155
Hail damage	84.02%	76.44%	80.05%	100.00%	100.00%	100.00%	399
Mixed Distress	80.27%	66.85%	72.95%	100.00%	99.72%	99.86%	359
Root Damage	77.27%	87.50%	82.07%	100.00%	100.00%	100.00%	136
Strong-Light Burn	53.25%	79.65%	63.83%	99.12%	100.00%	99.56%	113
Macro Average	72.07%	77.06%	73.76%	99.82%	99.94%	99.88%	1,162

Table 3: **Per-Class Evaluation Metrics on the Held-Out Test Set.** Note the massive precision recovery in Strong-Light Burn and Bollworm Infestation.

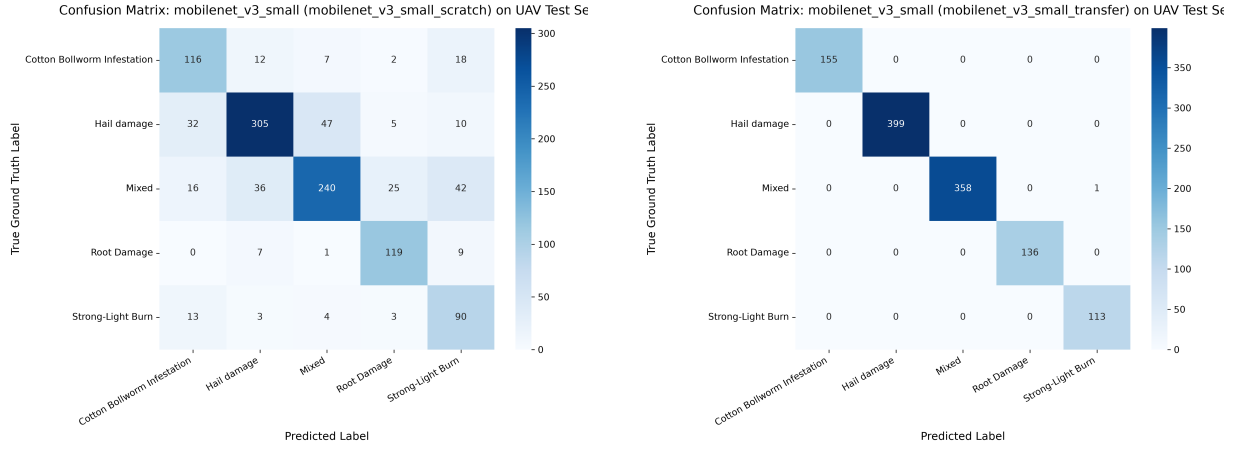
7.1 Real-Time Flight Envelope and Companion Compute

In commercial UAV agricultural missions, imagery is acquired at frame rates between 30 and 60 FPS from high-resolution gimbals. An effective onboard diagnostic payload must classify incoming frames in real-time without buffering backlogs:

- **Latency Envelope:** MobileNetV3-Small processes each frame in **6.15 ms** (≈ 162.7 FPS), occupying merely 37% of a standard 60 FPS frame budget (16.6 ms). This leaves ample compute for simultaneous visual odometry, SLAM, and GPS geotagging.
- **Embedded Low-Power Feasibility:** On low-power embedded companion computers (e.g., NVIDIA Jetson Orin Nano with a 7W–15W power envelope or Raspberry Pi 5 with an Edge TPU), MobileNetV3-Small’s 1.52M parameter footprint translates to negligible memory bandwidth stress and low thermal throttling risks.
- **The Inefficiency of Heavy Architectures:** EfficientNet-B3 requires 17.86 ms on desktop GPU hardware, which would exceed real-time 60 FPS thresholds on embedded edge hardware, with no discernible gain in diagnostic accuracy.

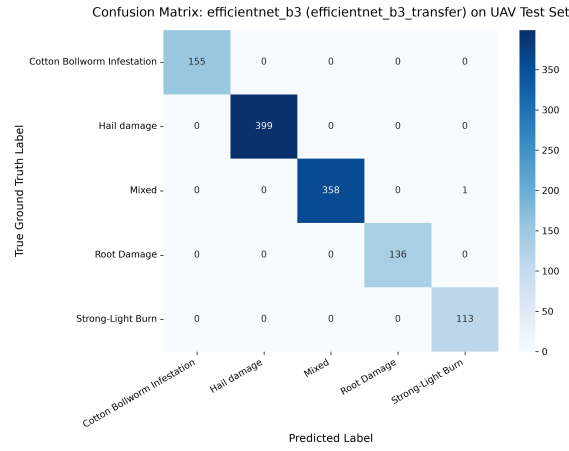
8 Conclusion

This investigation resolved the data leakage pitfalls of the Bangladesh UAV plant disease benchmark and demonstrated the decisive role of transfer learning in agricultural aerial diagnostics. Specifically:



(a) MobileNetV3-Small (From Scratch)

(b) MobileNetV3-Small (Transfer Learning)



(c) EfficientNet-B3 (Transfer Learning)

Figure 4: **Confusion Matrices on 1,162 Held-Out UAV Test Samples.** While the from-scratch model exhibits widespread off-diagonal dispersion, the transfer learning models achieve near-perfect diagonal concentration (1,161 out of 1,162 correct).

1. **Leakage Elimination:** Purged 1,040 cross-split duplicates, creating a clean 7,744-image benchmark with zero label ambiguity.
2. **Ablation Confirmation:** Demonstrated that transfer learning elevates a lightweight, mobile-optimized model (MobileNetV3-Small) from an inadequate 74.87% scratch baseline to a near-flawless **99.91% test accuracy**.
3. **Architectural Parity:** Proved that low-parameter models with rich inductive transfer can fully match the accuracy of architectures $7\times$ their size (EfficientNet-B3) while operating at **162.7 FPS**.

All model checkpoints, split manifests, and pipeline scripts are fully preserved in the **TrenUAV** repository for edge compilation and onboard UAV deployment.